

TRANSSERIES, ASYMPTOTIC INVERSION, AND CONDITIONAL PROBABILITY

The Lower Critical Endpoint

Arbitrary-Rate Condensation, Cauchy Cutoff Laws, and Degenerating Transseries

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Abstract

We study the critical countable-action equation

$$U_\epsilon(q) = c_\epsilon(\text{Li}_{2+\epsilon}(qe^{U_\epsilon(q)}) + P(qe^{U_\epsilon(q)})), \quad c_\epsilon = (\zeta(1+\epsilon) + P'(1))^{-1},$$

where P changes finitely many nonnegative action weights. For every joint limit $n \rightarrow \infty$, $\epsilon = \epsilon_n \downarrow 0$, with no restriction on its rate, we prove

$$[q^n]U_\epsilon(q) \sim \rho_\epsilon^{-n} \frac{\epsilon}{nd_n}, \quad d_n = \left(\frac{nc_\epsilon}{\epsilon}\right)^{1/(1+\epsilon)}, \quad \rho_\epsilon = e^{-c_\epsilon(\zeta(2+\epsilon)+P(1))}.$$

The coefficient is produced by one exceptional action. Deleting that action from the exactly conditioned Poisson configuration recovers the unconditioned configuration in total variation. If $nc_\epsilon \rightarrow \infty$, the remaining actions have a Cauchy-domain cloud on the scale $b_n = (nc_\epsilon)^{1/(1+\epsilon)}$, and the largest action has a reflected, explicitly normalized 1-stable fluctuation about an exact truncated-mean center D_n . Consequently, the minimal action cutoff retaining a fixed fraction r of the coefficient is $D_n + b_n F^{-1}(r) + o(b_n)$. Replacing D_n by d_n is incorrect at that precision: their difference is asymptotic to $b_n \log(1/\epsilon)$. The bounded-intensity regime has instead a discrete compound-Poisson remainder. We also give a convergent two-coordinate critical inverse chart, uniform through the removable parameter endpoint, and explain why its endpoint is a degenerate equation rather than a new nonzero critical function. All probabilistic estimates needed for the moving-parameter limits are proved directly.

Scope and status. This article supplies a model-specific answer to the lower-endpoint part of Question 4 in the repository's *Critical Hahn Transseries* report. Cauchy condensation and deletion of a largest summand are established mechanisms, not discoveries claimed here. The contribution is the explicit arbitrary-rate triangular theorem, its sparse/dense matching, and its transseries and action-budget consequences for this family. Publication priority has not been established. These are conventional mathematical proofs in an unrefereed research draft; no new Lean verification is claimed.

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1 The precise question and the contribution

1.1 A repository question not answered by the upper endpoint

The **ProveIt** transseries project contains a countable-action critical-inversion model with tail $a_j \asymp j^{-1-\alpha}$, $1 < \alpha < 2$. Its report *Beyond Finite-Action Folds: Critical Hahn Transseries, Stable Sector Asymptotics, and Sharp Action Budgets* asks in Question 4 for joint limits $\alpha \downarrow 1$ and $\alpha \uparrow 2$ with coefficient index n [1]. The editorial note at the snapshot inspected here records several results near the upper endpoint, but explicitly leaves the lower endpoint untreated. The snapshot is

3d5973524506411392a911470b5ddc35521568ea.

The relevant question occurs at source lines 1502–1507 of the critical Hahn article; the surrounding editorial note distinguishes the two endpoints. The separate transseries project and the intake ledger are documented in [2, 3].

At the upper endpoint the issue is a variance changing from infinite to finite. At the lower endpoint it is the mean that would cease to be finite without a simultaneous normalization. The critical normalization tends to zero. Thus a Gaussian argument with a modified variance cannot simply be reused, and a fixed- α stable local limit cannot be substituted into a moving- α problem without new estimates.

We resolve this lower-endpoint problem for an exact power tail with a fixed finite prefix. The class is broad enough to test universality under local changes while retaining explicit normalization and exact tail sums. We do not purport to cover arbitrary slowly varying tails, moving prefixes, or a simultaneous departure from critical coupling.

1.2 What belongs to the literature

The change of variable $t = qe^U$, Lagrange inversion, and conditioning identities belong to the standard allocation and simply generated tree framework; Janson gives a comprehensive treatment [4]. The principle that one large summand realizes a heavy-tailed deviation, and that deleting it can restore the original joint law, is developed by Armendáriz and Loulakis [5]. Critical Cauchy condensation, a separation between the largest component and the remaining extreme-value scale, and Cauchy fluctuations are already present in the work of Kortchemski and Richier [6]. Their setting is a fixed critical Cauchy-domain offspring law. Here the action law itself changes with n , its exponent approaches one from above, and its total intensity may tend to infinity, remain bounded, or vanish.

Accordingly, the new result claimed within this article is not the existence of condensation in general. It is the particular uniform triangular-array theorem proved below, including exact critical centering, finite-prefix dependence, and the resulting coefficient and cutoff laws. We do not invoke a fixed-law condensation theorem as though it were uniform in our degenerating parameter. The direct estimates are included precisely to close that logical gap. The special-function input is the classical polylogarithm expansion recorded in DLMF 25.12.12 [7].

1.3 A guide to the results

The main conclusions are collected in Theorem 3.1. The analytic chart is constructed in Section 4. Sections 5 and 6 prove the dense local theorem and deletion in total variation, without any rate restriction on ϵ_n . Section 7 handles bounded intensity, including intensity tending to zero. The Cauchy cloud, action cutoffs, and coefficient matching are established in Sections 8–10. Exact algebra checks and floating-point diagnostics are separated in Section 11.

The strongest reusable ingredient is the deletion theorem. It transfers *all bounded observables of the remaining configuration*, not just its largest member or its total weight. The distinction between the leading condensate scale d_n and the fluctuation center D_n is especially important for computation: asymptotically equivalent centers need not be interchangeable in a cutoff quantile.

2 Model, notation, and exact identities

2.1 The finite-prefix class

Fix a real polynomial

$$P(t) = \sum_{j=1}^J p_j t^j, \quad \kappa = P'(1) = \sum_{j=1}^J j p_j. \quad (2.1)$$

Set $p_j = 0$ for $j > J$. Assume that for all sufficiently small real $\epsilon > 0$,

$$j^{-2-\epsilon} + p_j \geq 0 \quad (j \geq 1), \quad 1 + p_1 > 0. \quad (2.2)$$

Negative p_j are allowed; it is the resulting weights that must be nonnegative. The polynomial P is fixed throughout every joint limit. Write

$$\alpha = 1 + \epsilon, \quad A_\epsilon(t) = \text{Li}_{2+\epsilon}(t) + P(t), \quad (2.3)$$

$$c_\epsilon = \frac{1}{\zeta(1 + \epsilon) + \kappa}, \quad m_\epsilon = c_\epsilon A_\epsilon(1), \quad \rho_\epsilon = e^{-m_\epsilon}. \quad (2.4)$$

For sufficiently small ϵ we have $c_\epsilon > 0$, and

$$c_\epsilon A'_\epsilon(1) = 1, \quad c_\epsilon = \epsilon - (\gamma + \kappa)\epsilon^2 + O(\epsilon^3). \quad (2.5)$$

Here γ denotes Euler's constant.

Define $U_\epsilon \in q\mathbb{R}[[q]]$ by

$$U_\epsilon(q) = c_\epsilon A_\epsilon(qe^{U_\epsilon(q)}), \quad u_n(\epsilon) = [q^n]U_\epsilon(q). \quad (2.6)$$

Each coefficient is determined recursively by lower coefficients. The analytic implicit function theorem at $(q, U) = (0, 0)$ also gives a convergent solution near the origin. Positivity and the positive first weight imply $u_n(\epsilon) > 0$ for every $n \geq 1$.

For an integer $M \geq 0$, let $A_{\epsilon, M}$ retain only the terms of A_ϵ with $j \leq M$, and define $U_{\epsilon, M}$ and $u_{n, M}$ using the *same* c_ϵ . Truncation does not renormalize the coupling. This convention is essential to the cutoff identity below.

2.2 Poisson configurations

For each pair (n, ϵ) , put

$$\lambda = nc_\epsilon, \quad b = \lambda^{1/\alpha}, \quad d = (\lambda/\epsilon)^{1/\alpha}. \quad (2.7)$$

Let the independent variables N_j have laws

$$N_j \sim \text{Pois}(\nu_j), \quad \nu_j = \lambda(j^{-1-\alpha} + p_j), \quad K = \sum_{j \geq 1} j N_j. \quad (2.8)$$

We extend $\nu_j = 0$ for integer $j \leq 0$. A restricted power multiplied by an event indicator is evaluated on that event and defined to be zero outside it. Since $\sum_j \nu_j < \infty$, the count vector

has finite support almost surely. Its total weight satisfies $\mathbb{E}K = n$. Conditioned on $K = n$, list the actions, with multiplicity, in nonincreasing order:

$$J_1 \geq J_2 \geq \dots \geq 0.$$

Zeros pad a finite list. Let $\mathcal{D}(N)$ be the count vector after removing one occurrence of a largest action; the choice between equal largest occurrences does not change the count vector.

We use $d_{TV}(\mu, \nu) = \sup_E |\mu(E) - \nu(E)|$ for total variation distance. The space of finitely supported count vectors on the positive integers is countable, so no additional measurable-space construction is required. All conditional assertions about J_i refer to $\mathbb{P}(\cdot \mid K = n)$.

Proposition 2.1 (Exact coefficient and cutoff identities). *With $t = qe^{U_\epsilon(q)}$,*

$$U_\epsilon = c_\epsilon A_\epsilon(t), \quad q = te^{-c_\epsilon A_\epsilon(t)}, \quad u_n(\epsilon) = \frac{1}{n} [t^n] e^{nc_\epsilon A_\epsilon(t)}. \quad (2.9)$$

Consequently,

$$u_n(\epsilon) = \frac{\rho_\epsilon^{-n}}{n} \mathbb{P}(K = n), \quad \frac{u_{n,M}(\epsilon)}{u_n(\epsilon)} = \mathbb{P}(J_1 \leq M \mid K = n). \quad (2.10)$$

The ratio is nondecreasing in M and is exactly one for $M \geq n$.

Proof. The first two identities are direct substitutions. Lagrange inversion for $t = q \exp(c_\epsilon A_\epsilon(t))$ gives

$$[q^n]_{c_\epsilon A_\epsilon(t(q))} = \frac{1}{n} [t^{n-1}]_{c_\epsilon A'_\epsilon(t)} e^{nc_\epsilon A_\epsilon(t)} = \frac{1}{n} [t^n] e^{nc_\epsilon A_\epsilon(t)}.$$

For the last equality, differentiate the exponential and compare its coefficient of t^{n-1} . The probability generating function of K is $\exp(nc_\epsilon(A_\epsilon(t) - A_\epsilon(1)))$, proving the first probability identity. Forbidding all actions above M multiplies the truncated Poisson probability by $\exp(-\sum_{j>M} \nu_j)$. Thus its generating function is $\exp(nc_\epsilon A_{\epsilon,M}(t) - nc_\epsilon A_\epsilon(1))$, proving the second identity. Monotonicity and exactness for $M \geq n$ follow from the event description. $\square$

3 Main theorem: arbitrary-rate approach to the lower endpoint

Throughout a limit means $n \rightarrow \infty$ through integers and $\epsilon = \epsilon_n \rightarrow 0^+$, unless a different limit is explicitly specified. Subscripts on b, d, λ are often suppressed. No monotonicity of ϵ_n is required.

Define the real random variable Z by its characteristic function

$$\begin{aligned} \mathbb{E}e^{itZ} &= \exp \left\{ \int_0^\infty (e^{itx} - 1 - itx \mathbf{1}_{x \leq 1}) \frac{dx}{x^2} \right\} \\ &= \exp \left\{ -\frac{\pi}{2} |t| + it(1 - \gamma - \log |t|) \right\} \quad (t \neq 0), \end{aligned} \quad (3.1)$$

with value one at $t = 0$. This is a totally right-skewed 1-stable law with the displayed location convention; it is not a symmetric Cauchy law. Write

$$F(x) = \mathbb{P}(-Z \leq x). \quad (3.2)$$

The function F is continuous and strictly increasing from zero to one, as proved in Section 8.

Theorem 3.1 (Uniform coefficient, deletion, and cutoff laws). *For the fixed-prefix family (2.1)–(2.6) the following statements hold.*

(a) **Every rate of approach.** For every $\epsilon_n \rightarrow 0^+$,

$$\boxed{u_n(\epsilon_n) \sim \rho_{\epsilon_n}^{-n} \frac{\epsilon_n}{n d_n}}. \quad (3.3)$$

In the conditioned configuration,

$$d_{\text{TV}}(\mathcal{L}(\mathcal{D}(N) \mid K = n), \mathcal{L}(N)) \longrightarrow 0, \quad \frac{J_1}{d_n} \xrightarrow{\mathbb{P}} 1, \quad \frac{J_2}{J_1} \xrightarrow{\mathbb{P}} 0. \quad (3.4)$$

(b) **Diverging intensity.** Suppose $\lambda_n = n c_{\epsilon_n} \rightarrow \infty$. Put

$$\mu_b = \sum_{j \leq b} j \nu_j, \quad D_n = n - \mu_b. \quad (3.5)$$

For all sufficiently large n ,

$$D_n = \lambda_n \zeta(1 + \epsilon_n, \lfloor b_n \rfloor + 1) = \frac{b_n}{\epsilon_n} + O(1). \quad (3.6)$$

Then

$$\frac{J_1 - D_n}{b_n} \xrightarrow{d} -Z, \quad \sum_{k \geq 2} \delta_{J_k/b_n} \xrightarrow{d} \Pi, \quad (3.7)$$

where Π is a Poisson point process on $(0, \infty)$ with intensity $x^{-2} dx$, with convergence of point measures away from zero. The two limits hold jointly, with Z the compensated sum of this same point process. In particular, for $x \in \mathbb{R}$ and $r \in (0, 1)$,

$$\frac{u_{n, \lfloor D_n + b_n x \rfloor}(\epsilon_n)}{u_n(\epsilon_n)} \longrightarrow F(x), \quad (3.8)$$

$$M_r(n) := \min\{M \in \mathbb{N} : u_{n,M}/u_n \geq r\} = D_n + b_n F^{-1}(r) + o(b_n). \quad (3.9)$$

(c) **Bounded intensity.** If $\lambda_n \rightarrow \lambda_0 \in [0, \infty)$, let independent $N_j^0 \sim \text{Pois}(\lambda_0(j^{-2} + p_j))$ and $S_{\lambda_0} = \sum_j j N_j^0$. Then

$$n - J_1 \xrightarrow{d} S_{\lambda_0}, \quad \mathbb{P}(K = n) \sim \lambda_n n^{-2-\epsilon_n}. \quad (3.10)$$

The first convergence holds in total variation as a convergence of integer-valued laws. If $\lambda_0 = 0$, $J_1 = n$ with probability tending to one.

Remark 3.2 (Meaning of arbitrary-rate uniformity). Part (a) holds along every pair sequence $n \rightarrow \infty$, $\epsilon \rightarrow 0^+$. Equivalently, for every $\eta > 0$ there are N and $\epsilon_0 > 0$ such that the relative error in (3.3) is below η whenever $n \geq N$ and $0 < \epsilon \leq \epsilon_0$. This equivalence is the elementary sequential characterization of a two-variable limit. It gives existence of uniform thresholds, not effective numerical values of those thresholds.

Remark 3.3 (What is and is not universal). In (b) the limiting standardized cloud is independent of the finite prefix. The normalizers $c_\epsilon, \rho_\epsilon, b_n, D_n$ still depend on it. In (c) the limiting discrete cloud retains all finite-prefix weights. Neither statement says that changing a prefix leaves raw coefficients asymptotically unchanged: the factor ρ_ϵ^{-n} can magnify a small change exponentially.

4 A convergent critical inverse chart through the degenerating endpoint

4.1 Exact phase and removable singularities

Take $t = e^{-x}$, with a fixed branch of $\log x$ near the positive real axis, and let $h = \log(\rho_\epsilon/q)$. The exact parametrization gives

$$h = x + c_\epsilon(A_\epsilon(e^{-x}) - A_\epsilon(1)), \quad U_\epsilon = m_\epsilon + h - x. \quad (4.1)$$

The polylogarithm expansion [7, Eq. 25.12.12] yields

$$h = B_\epsilon x^{1+\epsilon} + c_\epsilon x^2 R_\epsilon(x), \quad B_\epsilon = c_\epsilon \Gamma(-1-\epsilon), \quad (4.2)$$

where

$$R_\epsilon(x) = \sum_{k \geq 2} \frac{(-1)^k}{k!} \left(\zeta(2+\epsilon-k) + \sum_{j=1}^J p_j j^k \right) x^{k-2}. \quad (4.3)$$

The radius in x is at least 2π before choosing a smaller common polydisc. In particular the linear term in (4.1) vanishes *exactly*, by criticality.

Lemma 4.1 (Uniform analytic coefficients). *There is $e_0 > 0$ such that c_ϵ , B_ϵ , and $R_\epsilon(x)$ extend holomorphically in ϵ to $|\epsilon| < e_0$, with R jointly holomorphic for $|x| < r_0$ for some $r_0 > 0$. Moreover*

$$c_0 = 0, \quad B_0 = 1, \quad B_\epsilon = 1 - (1+\kappa)\epsilon + O(\epsilon^2). \quad (4.4)$$

After shrinking e_0 , B_ϵ has no zero there.

Proof. Write $\zeta(1+\epsilon) = \epsilon^{-1} + \gamma + O(\epsilon)$ and $\Gamma(-1-\epsilon) = \epsilon^{-1} + \gamma - 1 + O(\epsilon)$. The first reciprocal has a simple zero at zero and cancels the latter pole. Multiplication proves (4.4). The coefficients of (4.3) have no pole near $\epsilon = 0$. The convergent polylogarithm expansion, or its locally uniformly convergent power-series remainder, shows joint holomorphy in a smaller polydisc. Nonvanishing follows from $B_0 = 1$ and continuity. $\square$

4.2 Uniform two-coordinate inversion

Set $\delta = 1 - \epsilon = 2 - \alpha$ and

$$y = (h/B_\epsilon)^{1/\alpha}, \quad z = \frac{c_\epsilon}{B_\epsilon} y^\delta = \frac{y^\delta}{\Gamma(-1-\epsilon)}. \quad (4.5)$$

Branches are taken consistently with positive h, x, y for positive real arguments. The variables y, z in the next theorem are first independent complex coordinates; only afterwards are they specialized as above.

Theorem 4.2 (Uniform convergent inverse chart). *There are positive radii r, s, e_1 and a unique holomorphic function $V(y, z, \epsilon)$ near $\{|y| < r, |z| < s, |\epsilon| < e_1\}$, with $|V - 1| < 1/4$, satisfying*

$$V^\alpha(1 + zV^\delta R_\epsilon(yV)) = 1, \quad V(y, 0, \epsilon) = 1. \quad (4.6)$$

In these coordinates the local inverse is

$$x = yV(y, z, \epsilon), \quad U_\epsilon = m_\epsilon + h - yV(y, z, \epsilon). \quad (4.7)$$

The double power series of V converges absolutely on a common smaller polydisc. Thus, for each sufficiently small real $\epsilon > 0$, the specialized x is a convergent Hahn series on the positive grid

$$\{1 + k + m(1 - \epsilon) : k, m \in \mathbb{N}\}. \quad (4.8)$$

The coefficient functions extend holomorphically through $\epsilon = 0$.

Proof. At $V = 1, z = 0$ the left side of (4.6) minus one vanishes, and its V derivative is α , bounded away from zero for small complex ϵ . The holomorphic implicit function theorem supplies the chart, uniformly after restricting $|y|, |\epsilon|$ to smaller closed discs. Equivalently, consider

$$T(V) = (1 + zV^\delta R_\epsilon(yV))^{-1/\alpha}.$$

On $|V - 1| \leq 1/4$, small common radii bound R , its derivative, and the chosen powers. For sufficiently small s , T maps this closed disc into its interior and has derivative of modulus at most $1/2$. This proves uniqueness in the stated disc and gives a convergent iteration there. Substituting $x = yV$ into (4.2) gives exactly (4.6). Ordinary Cauchy estimates on a smaller polydisc prove absolute convergence of the double power series. Specialization of z then produces the grid (4.8). Its generators are strictly positive, so only finitely many pairs contribute below any fixed exponent bound. Coinciding exponents are simply collected. Holomorphic dependence on ϵ was part of the implicit-function construction. $\square$

Proposition 4.3 (First chart coefficients and a geometric error bound). *For fixed ϵ , write $R = R_\epsilon(y)$ and $R' = R'_\epsilon(y)$. Then*

$$V = 1 - \frac{R}{\alpha}z + \frac{(5 - \alpha)R^2 + 2yRR'}{2\alpha^2}z^2 + O(z^3), \quad (4.9)$$

uniformly on smaller (y, ϵ) polydiscs. Suppose $|V| \leq H$ on $|y| \leq r$, $|z| \leq s$ at a fixed admissible parameter. If $V_{K,M}$ retains $0 \leq k \leq K$, $0 \leq m \leq M$ in the double Taylor series, then for $a = |y|/r < 1$ and $v = |z|/s < 1$,

$$|V - V_{K,M}| \leq \frac{H(a^{K+1} + v^{M+1})}{(1 - a)(1 - v)}. \quad (4.10)$$

The bound is uniform whenever the same H, r, s work for the parameters.

Proof. Using $\alpha + \delta = 2$, rewrite (4.6) as $V^\alpha + zV^2R_\epsilon(yV) = 1$. Substitute $V = 1 + az + bz^2 + O(z^3)$. The coefficients of z and z^2 respectively give

$$\alpha a + R = 0, \quad \alpha b + \frac{\alpha(\alpha - 1)}{2}a^2 + a(2R + yR') = 0,$$

which prove (4.9). Cauchy's estimate gives $|[y^k z^m]V| \leq Hr^{-k}s^{-m}$. Sum the two geometric tails outside the retained rectangle; counting their intersection twice only increases the upper bound. $\square$

4.3 The endpoint is degenerate, not a hidden nonzero solution

At $\epsilon = 0$ we have $c_0 = 0$, $B_0 = 1$, and the specialized z is zero. Hence $h = x$, $m_0 = 0$, and (4.7) gives $U_0 = 0$. This does not contradict the nontrivial coefficient theorem: n grows while the parameter approaches its degenerate endpoint.

Proposition 4.4 (Fixed-domain limit). *On each closed disc $|q| \leq r < 1$,*

$$\frac{U_\epsilon(q)}{c_\epsilon} \longrightarrow \text{Li}_2(q) + P(q) \quad (4.11)$$

uniformly as $\epsilon \downarrow 0$.

Proof. Choose $r < r' < 1$. For $|q| \leq r$ and $|U|$ small enough, $|qe^U| \leq r'$. The functions A_ϵ and their derivatives are uniformly bounded on this latter disc and converge there uniformly to A_0 . For small c_ϵ , the map $U \mapsto c_\epsilon A_\epsilon(qe^U)$ is a uniform contraction on a disc of radius Cc_ϵ . Consequently $U_\epsilon = O(c_\epsilon)$ uniformly. Divide the equation by c_ϵ and pass to the uniform limit. $\square$

5 Uniform estimates for the Poisson array

This section supplies the local estimates that cannot be obtained merely by substituting a moving exponent into a fixed-parameter theorem. Constants may depend on the fixed prefix, but not on n or on sufficiently small ϵ . Eventually we restrict $1 < \alpha \leq 5/4$.

5.1 Tail sums and truncated variance

For $M > J$, define

$$K_M = \sum_{j \leq M} jN_j, \quad \Lambda_M = \sum_{j > M} \nu_j, \quad \Delta_M = n - \mathbb{E}K_M, \quad V_M = \text{Var } K_M. \quad (5.1)$$

The integral test gives, uniformly in the indicated range,

$$\Lambda_M = \frac{\lambda M^{-\alpha}}{\alpha} + O(\lambda M^{-1-\alpha}), \quad (5.2)$$

$$\Delta_M = \frac{\lambda M^{-\epsilon}}{\epsilon} + O(\lambda M^{-1-\epsilon}), \quad (5.3)$$

$$V_M \leq C\lambda(M^{2-\alpha} + 1), \quad \sup_{j > M} \nu_j \leq C\lambda M^{-1-\alpha}. \quad (5.4)$$

The mean identity uses $\mathbb{E}K = n$. Even when some p_j are negative, the variance is bounded above using $\sum_{j \leq J} |p_j|j^2$ plus the pure tail. If $b \rightarrow \infty$, (5.3) at $M = [b]$ gives

$$D = \lambda \sum_{j > [b]} j^{-1-\epsilon} = \frac{b}{\epsilon} + O(1), \quad (5.5)$$

because $\lambda = b^{1+\epsilon}$. The floor contributes only $O(1)$ as well.

Lemma 5.1 (Convolution bound for two or more large actions). *Let $\nu^>$ be the measure on integers with mass ν_j at $j > M$, let $\Lambda = \|\nu^>\|_1$, and let $H_M = \sum_{j > M} jN_j$. Then*

$$\sup_k \mathbb{P}(K_M + H_M = k, \text{ at least two actions exceed } M) \leq \|\nu^>\|_\infty \Lambda e^\Lambda. \quad (5.6)$$

Proof. The subprobability measure for exactly l large actions is $e^{-\Lambda}(\nu^>)^{*l}/l!$. For $l \geq 2$, $\|(\nu^>)^{*l}\|_\infty \leq \|\nu^>\|_\infty \Lambda^{l-1}$. Convolution with the probability law of K_M cannot increase the supremum norm. Sum over $l \geq 2$ and use $\sum_{l \geq 2} \Lambda^{l-1}/l! \leq \Lambda e^\Lambda$. $\square$

5.2 A local bound preserved by positive exponential tilting

Lemma 5.2 (Uniform lattice concentration). *Assume $b \rightarrow \infty$ and $M/b \rightarrow \infty$. For every $t \geq 0$, let $\widetilde{K}_M$ be compound Poisson with rates $\tilde{\nu}_j = \nu_j e^{tj}$, $j \leq M$. There is a constant C such that*

$$\sup_{k \in \mathbb{Z}} \mathbb{P}(\widetilde{K}_M = k) \leq \frac{C}{b} \quad (5.7)$$

for all sufficiently large n , uniformly in t .

Proof. Fourier inversion bounds the left side by

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} \exp \left(- \sum_{j \leq M} \nu_j e^{tj} (1 - \cos j\theta) \right) d\theta.$$

Since $t \geq 0$, it suffices to bound the untilted integral. The interval $|\theta| \leq b^{-1}$ contributes at most $2/b$. Choose a small fixed $\theta_0 > 0$ so that, for $b^{-1} \leq |\theta| \leq \theta_0$, the integers j in $[1/(2|\theta|), 1/|\theta|]$ include at least a constant multiple of $1/|\theta|$ indices above J . They are at most b and hence below M . On this range, $1 - \cos(j\theta)$ is bounded below by a positive constant. The exact tail gives

$$\sum_{j \leq M} \nu_j (1 - \cos j\theta) \geq c\lambda|\theta|^\alpha = c(b|\theta|)^\alpha.$$

The integral of this upper envelope is $O(1/b)$ uniformly for $1 \leq \alpha \leq 5/4$.

On $\theta_0 \leq |\theta| \leq \pi$, retain two consecutive fixed indices larger than J . Their trigonometric terms have no common zero in this compact set, and both rates are at least a positive constant times λ . The remaining integral is $O(e^{-c\lambda}) = O(1/b)$. All estimates survive the positive tilt. This proves the claim. $\square$

Lemma 5.3 (Tilt bound at a prescribed lattice point). *Under the hypotheses of Lemma 5.2, for $t \geq 0$,*

$$\mathbb{P}(K_M = n) \leq \frac{C}{b} \exp\left(-t\Delta_M + \frac{V_M}{M^2} e^{tM}\right). \quad (5.8)$$

Without the concentration hypothesis, the corresponding exponential factor bounds the tail probability $\mathbb{P}(K_M \geq n)$.

Proof. The exact exponential-tilt identity at n contributes the factor $\exp(-tn + \sum_{j \leq M} \nu_j (e^{tj} - 1))$ multiplying the tilted local probability. For $0 \leq j \leq M$, the power series of the exponential gives

$$e^{tj} - 1 - tj \leq \frac{j^2}{M^2} (e^{tM} - 1 - tM) \leq \frac{j^2}{M^2} e^{tM}.$$

Sum this inequality and use Lemma 5.2. The tail bound is the usual exponential Markov inequality with the same calculation. $\square$

6 Dense condensation and deletion in total variation

Assume throughout this section that $\lambda \rightarrow \infty$, equivalently $b \rightarrow \infty$. The proof does not require $\epsilon \log n$ to be bounded. Define the positive real tail weight

$$w_D = \lambda D^{-1-\alpha}. \quad (6.1)$$

This is evaluated at the real number D ; replacing it by the weight at $\lfloor D \rfloor$ has relative error tending to zero.

Theorem 6.1 (Dense local one-action theorem). *For every sequence with $\lambda \rightarrow \infty$ and $\epsilon \rightarrow 0^+$,*

$$\mathbb{P}(K = n) \sim w_D. \quad (6.2)$$

With conditional probability tending to one, exactly one action exceeds ηD , for any one fixed sufficiently small $\eta > 0$. That action, divided by D , tends to one in probability. Moreover the deletion conclusion (3.4) holds.

Proof. Fix $\eta = 1/64$ and take $M = \lfloor \eta D \rfloor$. Since $D = b/\epsilon + O(1)$,

$$\frac{M}{b} \rightarrow \infty, \quad \Lambda_M = O(\epsilon^\alpha) = o(1), \quad \sup_{j > M} \nu_j = O(w_D), \quad \frac{V_M}{D^2} = O(\epsilon^\alpha). \quad (6.3)$$

Furthermore, using (5.3),

$$\frac{\Delta_M}{D} = \left(\frac{b}{M}\right)^\epsilon (1 + o(1)) = \left(\frac{\epsilon}{\eta}\right)^\epsilon (1 + o(1)) \longrightarrow 1. \quad (6.4)$$

Chebyshev's inequality therefore gives

$$\frac{n - K_M}{D} \xrightarrow{\mathbb{P}} 1. \quad (6.5)$$

This statement concerns the unconditioned small-action configuration.

No large action. Set $tM = \frac{1}{2} \log(1/\epsilon)$. The positive remainder in the exponent of (5.8) is bounded by

$$C\lambda(M^{2-\alpha} + 1)M^{-2}\epsilon^{-1/2} = O(\epsilon^{\alpha-1/2}) + o(1) = o(1).$$

In the second term one may use $\lambda/M^2 = b^{\alpha-2}O(\epsilon^2)$ and $\alpha \leq 5/4$. Meanwhile $t\Delta_M = (1/(2\eta) + o(1)) \log(1/\epsilon)$. Thus for all sufficiently large n ,

$$\mathbb{P}(K_M = n) \leq \frac{C}{b} \epsilon^{16} = o(w_D), \quad (6.6)$$

because $w_D \sim \epsilon^{1+\alpha}/b$. The joint probability with no large action is still smaller. The factor $1/b$ in this estimate is crucial; a bound only on the upper tail would not suffice for every possible rate of growth of b .

At least two large actions. By Lemma 5.1 and (6.3), the local probability of $K = n$ with at least two actions above M is $O(w_D \Lambda_M) = o(w_D)$.

Exactly one large action. Its contribution is exactly

$$e^{-\Lambda_M} \mathbb{E}[\nu_{n-K_M} \mathbf{1}_{n-K_M > M}]. \quad (6.7)$$

For large n the prefix lies below M . Dividing the integrand by w_D gives a uniformly bounded nonnegative variable:

$$\frac{\nu_{n-K_M}}{w_D} \mathbf{1}_{n-K_M > M} = \left(\frac{n - K_M}{D}\right)^{-1-\alpha} \mathbf{1}_{n-K_M > M} \leq C_\eta.$$

By (6.5) it converges to one in probability and hence in L^1 . Since $\Lambda_M \rightarrow 0$, (6.7) is asymptotic to w_D . The three disjoint cases prove (6.2) and show that exactly one large action accounts for asymptotically all the conditioned probability. Its size divided by D tends to one by the same bounded weight argument.

Deletion. On the event just considered, the conditional law of the small count vector has density

$$\frac{\nu_{n-K_M} \mathbf{1}_{n-K_M > M}}{\mathbb{E}[\nu_{n-K_M} \mathbf{1}_{n-K_M > M}]} \quad (6.8)$$

relative to its original Poisson law. This density converges to one in L^1 , so the two laws converge in total variation. The original full configuration differs from the original small configuration with probability at most $1 - e^{-\Lambda_M} \rightarrow 0$, under their natural coupling. The exceptional conditioned events have probability tending to zero. Finally, on the exactly-one-large-action event the deleted action is the largest action. The triangle inequality for total variation proves the claim. $\square$

Remark 6.2 (Why critical mean does not prevent condensation). Every row has $\mathbb{E}K = n$, but its typical value is $\mu_b + O_{\mathbb{P}}(b)$, not n . Their difference is $D \sim b/\epsilon \gg b$. Very rare actions beyond the typical cloud carry the missing expectation. Conditioning on $K = n$ selects one such action. There is no contradiction between deletion in total variation and a changed expectation, since total variation controls bounded observables and these weights are not uniformly integrable. The fluctuation assertion used in this explanation is proved next in Section 8; it is not needed in the local proof above.

7 Bounded intensity and the discrete remainder

The regime $\lambda = nc_\epsilon = O(1)$ is not a small variation on a stable local limit: b need not diverge and can tend to zero. In particular a Fourier bound proportional to $1/b$ would be useless here. The following argument also keeps a factor of λ when $\lambda \rightarrow 0$, which is necessary for relative asymptotics.

Theorem 7.1 (Sparse local theorem). *Uniformly along sequences with $\lambda = O(1)$ and $\epsilon \rightarrow 0^+$,*

$$\mathbb{P}(K = n) \sim \lambda n^{-1-\alpha}. \quad (7.1)$$

Deletion of one largest action again gives the original Poisson configuration in total variation. If $\lambda \rightarrow \lambda_0$, the conclusion (3.10) follows.

Proof. Since $c_\epsilon/\epsilon \rightarrow 1$, bounded λ implies $\epsilon = O(1/n)$ and hence $\epsilon \log n \rightarrow 0$. Set $M = \lfloor \eta n \rfloor$ with $\eta = 1/64$. Then

$$\mathbb{E}K_M \leq C\lambda \log n, \quad V_M \leq C\lambda n, \quad \Lambda_M = O(\lambda/n), \quad \sup_{j>M} \nu_j = O(\lambda n^{-1-\alpha}). \quad (7.2)$$

In particular $K_M/n \xrightarrow{\mathbb{P}} 0$. Lemma 5.1 makes the local contribution of at least two large actions negligible relative to $\lambda n^{-1-\alpha}$. Exactly one large action contributes

$$e^{-\Lambda_M} \mathbb{E}[\nu_{n-K_M} \mathbf{1}_{n-K_M > M}] \sim \lambda n^{-1-\alpha},$$

by the same bounded-weight argument used in the dense case.

It remains to estimate zero large actions *relatively*, even when λ is arbitrarily small. Let $tM = \frac{1}{2} \log n$ and put $L = \sum_{j \leq M} \nu_j$, $L_t = \sum_{j \leq M} \nu_j e^{tj}$. The exponential estimate used before now shows

$$L_t - L \leq t\mathbb{E}K_M + \frac{V_M}{M^2} e^{tM} \leq C\lambda \frac{\log^2 n}{n} + C\lambda n^{-1/2} = o(1).$$

For $n > 0$, the zero configuration cannot contribute. Therefore

$$\begin{aligned} \mathbb{P}(K_M = n) &\leq e^{-tn} \mathbb{E}[e^{tK_M}; K_M > 0] \\ &= e^{-tn-L}(e^{L_t} - 1) \leq e^{-tn+L_t-L} L_t \leq C\lambda n^{-1/(2\eta)+1/2}. \end{aligned}$$

Here $L_t \leq e^{tM} L \leq C\lambda \sqrt{n}$. The final bound is $o(\lambda n^{-1-\alpha})$; we used $\epsilon \log n \rightarrow 0$ to compare $n^{-1-\alpha}$ with n^{-2} . This proves the local equivalent without dividing by an uncontrolled small probability.

The Radon–Nikodym density for the small configuration, conditioned on exactly one large action and total weight n , is again (6.8). Its normalization is now $\lambda n^{-1-\alpha}$. The density converges to one in L^1 , and $\Lambda_M \rightarrow 0$, proving convergence after deletion in total variation.

Finally, if $\lambda \rightarrow \lambda_0$, dominated convergence gives

$$\sum_{j \geq 1} \left| \lambda(j^{-2-\epsilon} + p_j) - \lambda_0(j^{-2} + p_j) \right| \rightarrow 0.$$

Independent Poisson configurations whose rate measures differ by this ℓ^1 amount can be coupled to agree except on an event of probability at most that amount, by using shared rates and independent excess rates. Thus the original full configuration converges in total variation to (N_j^0) . Deletion and the map $N \mapsto \sum j N_j$ preserve the resulting upper bound on total variation, and the total weight after deletion is exactly $n - J_1$. This proves the remaining assertions. $\square$

Remark 7.2 (Finite almost surely, infinite mean). For $\lambda_0 > 0$, the limiting sparse cloud has finite total number of actions almost surely because $\sum_j (j^{-2} + p_j) < \infty$. Its total weight is nevertheless unbounded and has infinite mean, since $\sum_j j(j^{-2} + p_j) = \infty$. There is no assertion of moment convergence in (3.10). When $\lambda_0 = 0$, its configuration is empty almost surely.

Corollary 7.3 (Discrete sparse cutoff law). *If $\lambda \rightarrow \lambda_0 \in [0, \infty)$, then for every fixed integer $k \geq 0$,*

$$\frac{u_{n,n-k}(\epsilon)}{u_n(\epsilon)} \longrightarrow \mathbb{P}(S_{\lambda_0} \geq k). \quad (7.3)$$

For $\lambda_0 = 0$ the limits are one for $k = 0$ and zero for each $k \geq 1$.

Proof. The event $J_1 \leq n - k$ is the event $n - J_1 \geq k$. Apply the integer-law total variation convergence from Theorem 7.1. $\square$

8 The Cauchy cloud and conditional extremes

Return to the dense regime $\lambda \rightarrow \infty$. The deletion theorem reduces the conditional limit laws to an unconditioned Poisson limit, which we now prove with the required moving exponent.

8.1 Joint convergence of points and compensated sums

Theorem 8.1 (Unconditioned triangular Cauchy limit). *If $b \rightarrow \infty$ and $\epsilon \rightarrow 0^+$, then*

$$\left(\sum_{j \geq 1} N_j \delta_{j/b}, \frac{K - \mu_b}{b} \right) \xrightarrow{d} (\Pi, Z), \quad (8.1)$$

where Π has intensity $x^{-2} dx$ and Z is its compensated sum:

$$Z = \lim_{a \downarrow 0} \left(\sum_{x \in \Pi, x > a} x - \log(1/a) \right). \quad (8.2)$$

The small-point compensation converges in L^2 ; the finitely many points above one are left uncompensated. The point process and Z are not independent.

Proof. For a continuous function f of compact support in $(0, \infty)$, eventually all indices in its support lie above the prefix. Since $\lambda = b^\alpha$,

$$\sum_j \nu_j f(j/b) = \frac{1}{b} \sum_j (j/b)^{-1-\alpha} f(j/b) \longrightarrow \int_0^\infty f(x) x^{-2} dx. \quad (8.3)$$

This is a Riemann sum, uniformly on compact intervals separated from zero, and $\alpha \rightarrow 1$. The Poisson Laplace functional $\exp(\sum_j \nu_j (e^{-f(j/b)} - 1))$ proves convergence of point measures on those intervals. In addition,

$$\mathbb{P}(\text{some action exceeds } Rb) \leq \sum_{j > Rb} \nu_j \leq CR^{-\alpha} \leq C/R \quad (R \geq 1) \quad (8.4)$$

for all sufficiently large n . This controls the passage to unbounded intervals at the upper end.

Fix $0 < a < 1$. The contribution from points above a to the centered sum is

$$\sum_{j > ab} \frac{j}{b} N_j - \sum_{ab < j \leq b} \frac{j}{b} \nu_j.$$

Together with the point process, it converges by (8.3) and (8.4) to $\sum_{x \in \Pi, x > a} x - \int_a^1 x^{-1} dx$. The error from the centered points below a has variance

$$\begin{aligned} \frac{1}{b^2} \sum_{j \leq ab} j^2 \nu_j &\leq C a^{2-\alpha} + C \lambda / b^2 \\ &\leq C a^{3/4} + o(1), \end{aligned}$$

where the second term bounds the fixed prefix and tends to zero because $\lambda/b^2 = b^{\alpha-2}$. This variance bound tends to zero after first letting $n \rightarrow \infty$ and then $a \downarrow 0$. The analogous limiting variance is $\int_0^a x^2 x^{-2} dx = a$; hence the compensated small-point sum converges in L^2 . Truncation at a followed by these variance estimates proves the joint convergence and the construction (8.2). Its characteristic function is precisely (3.1). $\square$

Proposition 8.2 (Normalization and regularity of the stable law). *The integral exponent in (3.1) equals $-\pi|t|/2 + it(1 - \gamma - \log|t|)$. The distribution of Z has a continuous real-analytic density, and its distribution function, and hence F , is strictly increasing.*

Proof. For real $s > 0$, differentiation under convergent truncated integrals gives

$$\frac{d}{ds} \int_0^\infty (e^{-sx} - 1 + sx \mathbf{1}_{x \leq 1}) \frac{dx}{x^2} = \log s + \gamma.$$

For example, the constant at $s = 1$ is the familiar integral $\int_0^1 (1 - e^{-x}) dx/x - \int_1^\infty e^{-x} dx/x = \gamma$; the derivative of the right side with respect to s is $1/s$. Integration from zero gives $s \log s + (\gamma - 1)s$. Analytic continuation to $s = -it$ from $\operatorname{Re} s > 0$, or direct dominated limits in the compensated integral, gives the stated Fourier exponent. The continuous extension at zero follows from $t \log|t| \rightarrow 0$.

The characteristic function has modulus $e^{-\pi|t|/2}$. Fourier inversion therefore supplies a continuous density that extends holomorphically to a strip of width less than $\pi/2$ about the real axis. It is nonnegative and has integral one. A real-analytic function that vanishes throughout an interval must vanish identically; consequently this density has positive integral on every nonempty interval. The distribution function is continuous and strictly increasing, with limits zero and one. $\square$

8.2 Transfer through deletion

Theorem 8.3 (Largest action and the remaining extremes). *In the dense regime, under exact conditioning $K = n$,*

$$\left(\sum_{k \geq 2} \delta_{J_k/b}, \frac{J_1 - D}{b} \right) \xrightarrow{d} (\Pi, -Z). \quad (8.5)$$

For every $x > 0$,

$$\mathbb{P}(J_2/b \leq x \mid K = n) \longrightarrow e^{-1/x}. \quad (8.6)$$

More generally, for each integer $k \geq 1$,

$$\mathbb{P}(J_{k+1}/b \leq x \mid K = n) \longrightarrow e^{-1/x} \sum_{l=0}^{k-1} \frac{x^{-l}}{l!}. \quad (8.7)$$

Proof. Total variation in Theorem 6.1 permits every measurable n -dependent mapping of the deleted configuration. Its point measure is the first component in (8.5). Its total weight is $n - J_1$, and

$$\frac{J_1 - D}{b} = -\frac{(n - J_1) - \mu_b}{b}.$$

Apply Theorem 8.1. To pass from point-process convergence to maxima, use (8.4) to control points above an arbitrarily large bound; the limiting process almost surely has no point at a prescribed x . The number of limiting points above x is Poisson with mean $\int_x^\infty y^{-2} dy = 1/x$. Zero such points gives (8.6), and at most $k - 1$ gives (8.7). $\square$

Corollary 8.4 (Separation of the condensate). *In the dense regime $J_1/D \xrightarrow{\mathbb{P}} 1$ and $J_2/J_1 \xrightarrow{\mathbb{P}} 0$. In the sparse regime $J_1/n \xrightarrow{\mathbb{P}} 1$ and again $J_2/J_1 \xrightarrow{\mathbb{P}} 0$.*

Proof. In the dense case, $b/D \sim \epsilon \rightarrow 0$, so tightness in (8.5) proves both claims. In the sparse case, bounded total intensity and the uniform bound $\sum_{j>R} \nu_j \leq C/R$ imply tightness of the unconditioned largest action. Deletion in total variation transfers this tightness to J_2 . Also $K_M/n \xrightarrow{\mathbb{P}} 0$ in the one-action proof gives $J_1/n \xrightarrow{\mathbb{P}} 1$. No assumption that λ has a limit is needed for these tightness statements. $\square$

9 Sharp action budgets and the logarithmic centering correction

9.1 From a conditional maximum to a coefficient cutoff

Theorem 9.1 (Cutoff profile and its quantiles). *If $\lambda \rightarrow \infty$, then (3.8) and (3.9) hold for every real x and every $r \in (0, 1)$.*

Proof. Combine the exact cutoff identity (2.10) with Theorem 8.3. The floor changes the standardized threshold by at most $1/b \rightarrow 0$, and F is continuous. The threshold lies in $[0, n]$ eventually for every fixed x : $D/b \rightarrow \infty$, and nonnegative rates and the exact tail imply

$$\frac{\mu_b}{b} \geq \sum_{j < j \leq b} \frac{1}{j} \rightarrow \infty,$$

since $b^\epsilon j^{-\epsilon} \geq 1$ for $j \leq b$. For a quantile, set $x_r = F^{-1}(r)$. For every $a > 0$, $F(x_r - a) < r < F(x_r + a)$. The corresponding cutoff ratios eventually bracket r , so monotonicity brackets $(M_r - D)/b$ between $x_r - a + o(1)$ and $x_r + a + o(1)$. Let $a \downarrow 0$. $\square$

The quantile law is asymptotic, not a finite- n certified error bound. In particular it does not license replacing F by a floating-point quadrature and calling the resulting integer cutoff a proof certificate.

9.2 Why the leading coefficient scale is the wrong cutoff center

Theorem 9.2 (Logarithmically many fluctuation widths). *In the dense regime,*

$$\frac{D}{d} \rightarrow 1, \tag{9.1}$$

$$\frac{D - d}{b} = \frac{\log(1/\epsilon)}{1 + \epsilon} + O(\epsilon \log^2(1/\epsilon)) + O(1/b). \tag{9.2}$$

Thus the quantile formula has the explicit form

$$\boxed{M_r(n) = d_n + b_n \left(\frac{\log(1/\epsilon_n)}{1 + \epsilon_n} + F^{-1}(r) \right) + o(b_n).} \tag{9.3}$$

For every fixed real C ,

$$\frac{u_{n, \lfloor d_n + C b_n \rfloor}(\epsilon_n)}{u_n(\epsilon_n)} \rightarrow 0. \tag{9.4}$$

Proof. From $D = b/\epsilon + O(1)$ and $d = b\epsilon^{-1/\alpha}$,

$$\frac{D}{d} = \epsilon^{-\epsilon/(1+\epsilon)} + O(1/d) \rightarrow 1.$$

Furthermore

$$\frac{D-d}{b} = \frac{1}{\epsilon} \left(1 - \exp \left(-\frac{\epsilon}{1+\epsilon} \log(1/\epsilon) \right) \right) + O(1/b).$$

Taylor's formula for the ordinary exponential gives (9.2). The error there tends to zero, so Theorem 9.1 gives (9.3). For the last assertion, $(d + Cb - D)/b \rightarrow -\infty$. Tightness of $(J_1 - D)/b$ and the cutoff identity then imply (9.4). $\square$

Remark 9.3 (A strong failure of a leading-order prescription). Both d and D are correct leading scales for the largest action. Nevertheless a cutoff equal to d plus any fixed number of cloud widths retains an asymptotically vanishing fraction of the coefficient. This is not a small error in a constant; it changes a desired fixed-retention budget into a zero-retention prescription. The logarithm in (9.3) is therefore an essential part of the endpoint answer.

9.3 The macroscopic crossover

Corollary 9.4 (Condensate fraction at every logarithmic rate). *If $\epsilon_n \log n \rightarrow \tau \in [0, \infty]$, then*

$$\frac{J_1}{n} \xrightarrow{\mathbb{P}} e^{-\tau}, \quad (9.5)$$

where $e^{-\infty} = 0$. For any arbitrary-rate sequence and any cutoffs with $M_n/d_n \rightarrow l \neq 1$,

$$\frac{u_{n,M_n}}{u_n} \rightarrow \begin{cases} 0, & 0 \leq l < 1, \\ 1, & l > 1. \end{cases} \quad (9.6)$$

At $l = 1$ no universal retained fraction is determined by this leading ratio alone.

Proof. Directly from (2.7),

$$\log(d/n) = -\frac{\epsilon}{1+\epsilon} \log n + \frac{1}{1+\epsilon} \log(c_\epsilon/\epsilon). \quad (9.7)$$

The last term tends to zero. The dense and sparse theorems give $J_1/d \xrightarrow{\mathbb{P}} 1$; for a sequence crossing between the two regimes, apply the subsequence argument in Section 10. Equation (9.7) proves (9.5), and convergence in probability gives (9.6). Equations (3.8) and (9.4) exhibit different limits with $M_n/d_n \rightarrow 1$. $\square$

Example 9.5 (Three concrete joint limits). For $\epsilon_n = n^{-1/2}$, $\lambda \sim \sqrt{n} \rightarrow \infty$ and the largest action occupies asymptotically all the mass, while its cloud has scale $b \sim \sqrt{n}$. A logarithmic budget shift of order $\sqrt{n} \log n$ is still present between d and the correct center.

For $\epsilon_n = \tau/\log n$ with $0 < \tau < \infty$,

$$d \sim e^{-\tau} n, \quad b \sim \frac{\tau e^{-\tau} n}{\log n}, \quad D - d \sim \frac{\tau e^{-\tau} n \log \log n}{\log n}.$$

Every row is exactly mean-critical, but the conditioned largest action carries the positive fraction $e^{-\tau}$.

For $\epsilon_n = (\log n)^{-1/2}$, the largest action is sublinear in n , but still asymptotically unique on its own scale d . It remains much larger than the next action: $d/b = \epsilon^{-1/(1+\epsilon)} \rightarrow \infty$.

10 Matching the coefficient laws and the critical transseries

10.1 The all-rate coefficient theorem

Proof of Theorem 3.1. First suppose $\lambda \rightarrow \infty$. Theorem 6.1 gives

$$\mathbb{P}(K = n) \sim \lambda D^{-1-\alpha} \sim \frac{\epsilon^{1+\alpha}}{b}.$$

The proposed universal expression is

$$\frac{\epsilon}{d} = \frac{\epsilon^{1+1/\alpha}}{b}.$$

Their ratio is $\epsilon^{\alpha-1/\alpha} \rightarrow 1$, because $(\alpha - 1/\alpha) \log \epsilon \rightarrow 0$. Combine this with Proposition 2.1.

If $\lambda = O(1)$, then $\epsilon \log n \rightarrow 0$, $c_\epsilon/\epsilon \rightarrow 1$, and $d/n \rightarrow 1$. Theorem 7.1 gives

$$\frac{\mathbb{P}(K = n)}{\epsilon/d} \sim \frac{nc_\epsilon n^{-2-\epsilon}}{\epsilon/d} = \frac{c_\epsilon}{\epsilon} \frac{d}{n} n^{-\epsilon} \rightarrow 1.$$

Deletion and separation have already been proved in both regimes.

For an arbitrary sequence, every subsequence has a further subsequence on which λ is bounded or tends to infinity: if it has a bounded subsequence use it; otherwise select a subsequence diverging to infinity. If any of the claimed scalar errors or total variation errors failed to tend to zero, select a subsequence on which it stays bounded away from zero and apply this dichotomy to obtain a contradiction. The same argument applies to convergence-in-probability error probabilities. Thus part (a) holds without restrictions on λ . Part (b) is Theorems 8.1, 8.3, and 9.1, with (5.5). Part (c) is Theorem 7.1. $\square$

10.2 The familiar singular coefficient survives, but needs a new proof

Set $p = 1/(1 + \epsilon)$ and recall $B_\epsilon = c_\epsilon \Gamma(-1 - \epsilon)$. The leading singular term in (4.7) is $-B_\epsilon^{-p} h^p$. Its familiar coefficient expression is

$$H_n(\epsilon) = \rho_\epsilon^{-n} \frac{B_\epsilon^{-p}}{-\Gamma(-p)} n^{-1-p}. \quad (10.1)$$

It is positive for $0 < p < 1$. A fixed- p singularity-transfer heuristic suggests this expression, but is not a uniform argument as $p \rightarrow 1$: the reciprocal gamma factor vanishes at that endpoint.

Corollary 10.1 (Uniform recovery of the singular coefficient). *For every sequence $n \rightarrow \infty$, $\epsilon_n \rightarrow 0^+$,*

$$u_n(\epsilon_n) \sim H_n(\epsilon_n). \quad (10.2)$$

More precisely the ratio of H_n to the elementary right side of (3.3) depends only on ϵ :

$$C(\epsilon) = (\epsilon \Gamma(-1 - \epsilon))^{-1/(1+\epsilon)} \frac{-1}{\epsilon \Gamma(-1/(1 + \epsilon))} = 1 - \epsilon + O(\epsilon^2). \quad (10.3)$$

Proof. Cancellation of the powers of n, c_ϵ, ϵ gives exactly (10.3). The first factor tends to one. For the second, $1 - p = \epsilon/(1 + \epsilon)$ and $-1/\Gamma(-p) \sim 1 - p \sim \epsilon$, so it also tends to one. Combining with Theorem 3.1 proves (10.2). For the displayed expansion, use

$$\epsilon \Gamma(-1 - \epsilon) = 1 + (\gamma - 1)\epsilon + O(\epsilon^2),$$

and, for $v = 1 - p$, $-1/\Gamma(-1 + v) = v + (\gamma - 1)v^2 + O(v^3)$. The two first-order factors are $1 - (\gamma - 1)\epsilon + O(\epsilon^2)$ and $1 + (\gamma - 2)\epsilon + O(\epsilon^2)$, whose product is as stated. $\square$

This corollary establishes a uniform coefficient statement without needing a uniform global complex contour or an unproved cancellation of all analytic terms at the moving endpoint. The local chart and the probability proof describe the same leading coefficient by independent routes. The corollary does *not* give a relative error of order $o(\epsilon)$ for the true coefficients; (10.3) is an exact comparison of two candidate leading expressions, not a bound on their remaining errors.

Remark 10.2 (Keep the exponential normalizer exact). The factor $\rho_\epsilon^{-n} = \exp(nc_\epsilon A_\epsilon(1))$ is exact in all displayed coefficient equivalents. Replacing its exponent by only the first term of its small- ϵ expansion generally leaves an error of order $n\epsilon^2$ in the exponent. That error need not tend to zero in an arbitrary-rate limit. The uniform theorem therefore does not authorize such a replacement without an additional rate assumption. The same care is required when a leading scale is used as a fluctuation center.

10.3 Matching sparse coefficients

If $\lambda_n \rightarrow \lambda_0 \in (0, \infty)$, then

$$u_n(\epsilon_n) \sim e^{\lambda_0(\zeta(2)+P(1))} \frac{\lambda_0}{n^3}. \quad (10.4)$$

If $\lambda_n \rightarrow 0$, the useful relative formula is instead $u_n \sim \epsilon_n/n^2$. These follow from the sparse local theorem, since $\rho_\epsilon^{-n} = \exp(\lambda A_\epsilon(1))$. The endpoint coefficient for fixed n from Proposition 4.4 is consistent with this very sparse regime, but it does not by itself justify letting n grow.

10.4 Why no contradiction with fixed-exponent behavior arises

Fixing $\epsilon > 0$ and letting $n \rightarrow \infty$ is not a sequence covered by the endpoint theorem. In particular the assertion that all but one action recover an unconditioned configuration in total variation is an endpoint statement, not a claim for every fixed $1 < \alpha < 2$. The parameter $\epsilon \rightarrow 0$ makes the exceptional action larger than the ordinary cloud by an unbounded factor. Similarly, finite- ϵ numerical ratios need not approach one when compared with the elementary constant in (3.3); their limiting constant can differ by $C(\epsilon)$. This distinction is visible in the diagnostics below.

11 Computation, exact checks, and reproducibility

The archive contains `verify.py`, its recorded `verification_results.json`, and the source of this article. The checks are supplementary; no proof above relies on extrapolating a numerical table.

11.1 Exact algebra

The script uses rational arithmetic in three different finite weight models. It computes the formal solution of the feedback equation through degree twelve and checks each coefficient against Lagrange inversion. It independently enumerates multiplicity vectors through degree eight and checks the exponential coefficient formula. These are 60 exact rational identities. Two further symbolic identities check the first two equations for the chart coefficients in Proposition 4.3. All 62 exact assertions passed in the recorded run.

These tests validate the finite coefficient bookkeeping and the displayed chart expansion. They are not a finite substitute for the local probability estimates, which concern unbounded indices and moving parameters.

11.2 Positive coefficient recurrence

For the numerical model, a useful recurrence avoids cancellation. If $p_k = \mathbb{P}(K = k)$, then

$$p_0 = e^{-\sum_j \nu_j}, \quad p_k = \frac{1}{k} \sum_{j=1}^k j \nu_j p_{k-j}. \quad (11.1)$$

This follows by differentiating the probability generating function. To compute the joint probability of $K = n$ and no action above M , set the rates above M to zero in the recurrence but retain the *original* p_0 . That convention preserves the probability of excluding the tail, so division by the untruncated p_n gives the exact mathematical cutoff ratio. The floating implementation is not exact arithmetic.

The recorded run uses extended-precision `numpy.longdouble` recurrences, with 70-digit `mpmath` normalization and chart calculations. It checks that the initial and final probabilities are positive and finite. Different platforms may provide a narrower `longdouble`; underflow raises an error rather than silently producing a result. The JSON records the mantissa precision.

Table 1: Coefficient diagnostics for $P = 0$. The elementary ratio is $p_n/(\epsilon/d)$; the gamma ratio uses $n\rho^n H_n$. Holding ϵ fixed is not the joint limit in the theorem.

n	ϵ	Elementary ratio	Gamma ratio
100	0.100	0.88565664	0.99693488
1500	0.100	0.88808230	0.99966531
4000	0.100	0.88824635	0.99984997
100	0.020	0.97903537	0.99964242
1500	0.020	0.97935966	0.99997353
4000	0.020	0.97937548	0.99998969
100	0.005	0.99488134	0.99992138
1500	0.005	0.99495423	0.99999463
4000	0.005	0.99495755	0.99999797

Table 2: Genuine triangular diagnostics: $\epsilon_n = n^{-1/2}$ and a sparse sequence $\epsilon_n = 1/n$. All entries use $P = 0$ and the elementary normalization in the main theorem.

n	$\epsilon_n = n^{-1/2}$	Dense ratio	Sparse ratio
100	0.100000	0.88565664	0.98967840
500	0.044721	0.95224627	0.99798733
1500	0.025820	0.97314044	0.99933193
4000	0.015811	0.98379152	0.99974980

11.3 Stable quantiles and chart checks

For the reflected law the inversion formula is

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty e^{-\pi t/2} \frac{\sin(t(x+1-\gamma-\log t))}{t} dt. \quad (11.2)$$

The apparent singularity at zero is integrable. The code uses numerical quadrature on a finite interval with an exponentially small omitted tail; its reported integration error is the quadrature routine's estimate, not a rigorous enclosure. Representative values and finite- n cutoff ratios are

Table 3: Cutoff diagnostics, $P = 0$, $\epsilon = 0.02$, and $M = \lfloor D + bx \rfloor$. Fixed- ϵ bias remains; these values are not finite- n error certificates.

n	x	M	Retained fraction	Limit $F(x)$
1500	-4	1276	0.351031	0.296865
1500	-2	1332	0.526884	0.472186
1500	0	1387	0.817415	0.789298
1500	1	1415	0.951970	0.942065
4000	-4	3339	0.351476	0.296865
4000	-2	3484	0.525311	0.472186
4000	0	3629	0.817311	0.789298
4000	1	3702	0.951051	0.942065

in Table 3. At these moderate parameters a visible finite- ϵ bias remains. The limiting theorem assumes both $\epsilon \rightarrow 0$ and $b \rightarrow \infty$, rather than $n \rightarrow \infty$ with ϵ fixed.

For $\epsilon \in \{0.2, 0.05, 0.001\}$ and $x \in \{0.01, 0.001\}$, the program checks the exact phase against a 30-term analytic remainder and reconstructs x by the contraction map in the proof of Theorem 4.2. The tested phase discrepancies are below 10^{-60} and the reconstructed discrepancies below 10^{-55} in 70-digit arithmetic. Those thresholds describe the computation, not a certified analytic remainder bound; the general bound is (4.10).

11.4 What has not been computationally certified

No interval-arithmetic cutoff algorithm is supplied. No numerical onset for the asymptotic equivalences is claimed. No floating-point table is called a proof of total variation convergence. No Lean file accompanies this article. The package separates a reproducible mathematical proof, exact finite algebra checks, and diagnostic floating calculations.

12 A formalization plan with explicit proof boundaries

The repository already has support and asymptotic-scale infrastructure [2, 3]. One relevant module is `TransseriesWellBased.lean`. That infrastructure should not be mistaken for a formal proof of the new probability estimates. The natural formalization is modular.

Finite algebra and exact coefficients. First prove the exponential-series recurrence over a rational algebra and its multiplicity-vector formula. Connect the formal implicit solution to Lagrange inversion. The positivity and finite-cutoff monotonicity statements can then be developed without analytic limits. A certificate checker for a finite list of rational weights should sit above these proved identities, not replace them.

Finite and countable Poisson configurations. Construct finite-support count vectors under a summable rate sequence, prove the generating-function identity, and connect the cutoff event to the coefficient ratio. The convolution estimate for multiple large actions is an ℓ^1 - ℓ^∞ inequality and is a useful early formal target. The deletion map is elementary on the count-vector space.

Uniform local control. Formalize the integral-test bounds with explicit constants on $1 \leq \alpha \leq 5/4$, then the trigonometric lower bound and Fourier concentration estimate. The exponential-

tilt identity should expose its hypotheses, particularly positivity of the tilt and the finite cutoff. Only then assemble the zero/one/multiple-action proof. The exact L^1 density argument is the central bridge from scalar estimates to total variation.

Moving-parameter convergence. A reusable Riemann-sum theorem with a parameter in a compact interval, a small-jump variance estimate, and a truncation lemma for joint weak convergence would support Theorem 8.1. Quantile convergence requires a separate continuity and strict-monotonicity interface; it is not an automatic consequence of pointwise coefficient asymptotics.

Analytic chart. The required ingredients are the polylogarithm expansion, cancellation of meromorphic poles in the parameter, and a parameter-uniform holomorphic implicit-function or contraction theorem. Ordinary multivariable power series can be used before constructing the specialized Hahn support. This keeps analytic existence distinct from formal support manipulation.

This is a plan, not a claim that these interfaces are already implemented in the pinned repository. In particular the present article adds no formal declarations to that repository and makes no changes to its files.

13 Further research questions

The results above answer the exact-tail lower-endpoint question at leading coefficient order and at the sharp fluctuation and cutoff scale. They also make the remaining questions more specific.

1. Effective error bounds and certified action selection. The proof gives explicit inequalities before several asymptotic simplifications. Optimize the splitting threshold $M = \eta D$, retain constants in the Fourier concentration bound, and obtain a computable upper bound on the deletion distance. Combine that bound with rigorous enclosures of (11.2) to return a certified finite- n cutoff for a prescribed retained fraction. The difficult step is not numerical integration alone; it is a quantitative comparison between the finite conditional law and its limit.

2. A uniform second-order coefficient theorem. Equation (10.3) compares two leading normalizations but does not bound the remainder by $o(\epsilon)$. Determine a joint expansion in ϵ , $1/b$, and the logarithmic combinations generated by $\epsilon \log \epsilon$. A useful target is an error formula valid across $\lambda = O(1)$, $\lambda \rightarrow \infty$ slowly, and very large b , rather than three unrelated expansions with no common overlap.

3. Slowly varying and nonexact tails. Replace $j^{-2-\epsilon}$ by $j^{-2-\epsilon} L_\epsilon(j)$. Identify uniform regularity conditions on L_ϵ that preserve one-action condensation, and construct the correct cloud and mean-defect scales. Ordinary regular variation for each fixed parameter need not be uniform as the parameter moves. Determine when a de Bruijn conjugate or a more general implicit scale replaces the explicit powers b and d .

4. Simultaneous coupling displacement. Allow the mean of K to be $n(1 - \delta_n)$ rather than n while $\epsilon_n \rightarrow 0$. Find the coupling window in which the imposed deficit, truncated-mean defect, and Cauchy cloud all interact. The natural centering candidate is $n - \sum_{j \leq b} j \nu_j$, but

positivity of that quantity and local one-action dominance must be re-established. The present proof does not address the side on which conditioning lies below the typical cloud.

5. A growing prefix and the loss of universality. Let the altered prefix length J_n grow. Determine sharp conditions on its centered variance, maximum action, and tail intensity under which it is still invisible after scaling by b_n . A prefix with negligible total mass can nevertheless shift a fluctuation center if its first moment is large. The fixed-prefix proof isolates exactly where stronger conditions would be required.

6. Several action species and competing condensates. For a sum of several near-index-one tails, mark the species of each action. Determine when exactly one condensate still occurs and how its species is selected. If different exponents approach one at different rates, the largest-action weight at the mean defect may select one species while the ordinary cloud contains several. Prove the corresponding marked point-process and cutoff statements before assuming a single effective exponent.

7. Geometry beyond the coefficient configuration. The conditioned Poisson configuration encodes coefficient contributions, not by itself a complete tree metric. Use an explicit combinatorial bijection to connect this triangular model to tree degree sequences, and then investigate height, diameter, and scaling limits. The fixed-law Cauchy-tree literature [6] supplies relevant comparisons, but its geometric conclusions cannot be transferred without that additional structure.

8. Complex continuation and additional transseries structure. Theorem 4.2 is a convergent local critical chart. It does not establish global analytic continuation, resurgence, Borel summability, or Stokes constants. Determine what happens when h and ϵ vary in larger complex domains, when the chosen branch winds, or when the parameter approaches other singularities of the normalization. State separately which properties follow from a finite-coordinate analytic chart and which require new global information.

9. Machine-checked deletion and triangular limits. Formalize the deletion theorem as a reusable interface: a negligible zero-large-action term, a negligible multi-large-action term, and an L^1 -asymptotically constant one-action weight imply total variation recovery of the cloud. This abstract statement would support many condensation models beyond the explicit zeta family, while leaving the model-specific tail and concentration estimates as separate obligations.

14 Conclusion

The lower endpoint does not merely replace one stable exponent by another. It turns the vanishing critical coupling into a mean-defect problem, with one exceptional action and a smaller compensated Cauchy cloud. For the exact power tail with a fixed finite prefix, the coefficient equivalent holds at every rate of approach, including regimes in which the total Poisson intensity stays bounded or tends to zero. In the dense regime the cutoff quantiles require a logarithmic correction to the leading scale; omitting it gives asymptotically zero retention even though the cutoff is correct to relative leading order.

The analytic and probabilistic descriptions fit together without a hidden uniformity assumption. A convergent critical chart persists through a removable parameter singularity, while a separate local probability proof establishes the uniform coefficient law. The matching gamma

expression is therefore a conclusion, not a substitution-based justification. The remaining tasks concern quantitative certification, less rigid tails, additional moving parameters, and formal verification.

A Proof-dependency and scope ledger

Statement	Proof inputs and boundary
Exact coefficients and cut-offs	Formal Lagrange inversion and summable Poisson rates; reproved in Proposition 2.1. No asymptotic assumption.
Uniform inverse chart	Classical polylogarithm expansion, removable pole cancellation, and a local holomorphic implicit-function argument. Local branches only.
Dense local equivalent	Integral-test estimates, a tilted Fourier concentration bound, and a decomposition into zero, one, or multiple large actions. No fixed-exponent local limit is imported.
Sparse local equivalent	A positive exponential-moment bound retaining a factor of λ , plus the same one-action decomposition. Valid when λ tends to zero.
Deletion in total variation	An explicit bounded Radon–Nikodym weight converging to one in L^1 ; no moment convergence asserted.
Cauchy cloud	Poisson Laplace functionals, compact Riemann sums, uniform upper-tail bounds, and small-jump variance control. Requires $b \rightarrow \infty$.
Sharp action budgets	Exact cutoff identity, deletion, weak convergence, and strict continuity of the limiting quantile. Not a finite- n certificate.
All-rate coefficient law	Dense and sparse local equivalents joined by a subsequence dichotomy; fixed prefix throughout.
Publication novelty	Not established. The arbitrary-rate theorem is presented as a proved, scoped contribution; classical condensation mechanisms are credited.
Lean status	No new Lean proofs or repository modifications are included.

B Reproduction instructions and source provenance

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